

Discovery of Temporal and Structural Patterns in U.S. Auto Sales Data

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Abstract—Government economic datasets offer rich opportunities for discovering meaningful patterns in large-scale human and market behavior. This project proposes a data mining study of U.S. auto sales data published by the Bureau of Transportation Statistics. The goal is to uncover recurring temporal patterns, structural groupings of sales periods, and anomalous deviations using unsupervised data mining techniques. The project emphasizes pattern discovery and interpretation rather than prediction, aligning with the Knowledge Discovery in Databases (KDD) process.

I. DATASET DESCRIPTION

The dataset used in this project is the *Auto Sales* dataset published by the U.S. Bureau of Transportation Statistics (BTS) and distributed through the Data.gov open data portal. The dataset contains historical statistics on automobile sales in the United States and is used in estimating national economic indicators such as personal consumption expenditures.

A. Source

- Bureau of Transportation Statistics (U.S. Department of Transportation)
- URL: <https://data.transportation.gov/d/7n6a-n5t>

B. Size and Structure

The dataset consists of time-series records of auto sales statistics. Depending on the selected time range, it contains:

- Hundreds to thousands of records
- Approximately 5–10 attributes
- Structured tabular format (CSV)

C. Key Attributes

The primary attributes used in this project include:

- Time period (year and/or month)
- Auto sales volume
- Vehicle category
- Aggregate sales indicators

D. Data Quality Issues

Potential data quality issues include missing values for certain time periods, inconsistent reporting intervals, and limited categorical diversity. These issues necessitate preprocessing and make the dataset well suited for anomaly detection and exploratory pattern analysis.

II. DISCOVERY QUESTIONS

This project is guided by the following discovery-focused questions:

- 1) What recurring temporal and structural patterns exist in U.S. auto sales, and how do these patterns vary across different economic periods?
- 2) Are there natural groupings of time periods that exhibit similar auto sales behavior?
- 3) Which time periods represent anomalous deviations from typical auto sales patterns, and what characteristics distinguish them?

These questions aim to uncover latent structure and meaningful trends rather than predict future sales values.

III. PLANNED TECHNIQUES

A. Clustering

Clustering techniques such as K-Means and hierarchical clustering will be applied to group time periods based on sales magnitude, growth rate, and trend features. Resulting clusters will be interpreted as distinct sales regimes or economic phases.

B. Association Rule Mining

Association rule mining (Apriori or FP-Growth) will be applied to discretized sales levels and temporal categories to discover frequent co-occurring patterns across time.

C. Anomaly Detection

Statistical and distance-based anomaly detection methods will be used to identify periods with unusually high or low sales compared to historical norms.

D. Dimensionality Reduction

Principal Component Analysis (PCA) will be used to reduce feature dimensionality, support visualization, and aid in understanding dominant sources of variance.

E. Analysis Pipeline

This project follows the Knowledge Discovery in Databases (KDD) process, progressing from data selection to interpretation of discovered patterns.

IV. PRELIMINARY TIMELINE

A. Milestone 2

Data acquisition, cleaning, handling missing values, normalization of sales metrics, feature engineering, and exploratory data analysis.

B. Milestone 3

Application of clustering, association rule mining, PCA, and anomaly detection techniques. Comparison of results across methods.

C. Milestone 4

Interpretation of discovered patterns, validation of findings, visualization, critical assessment of limitations, and preparation of the final report.

D. Anticipated Challenges

Challenges include selecting meaningful discretization thresholds, interpreting economic patterns without predictive assumptions, and assessing whether discovered patterns are robust or potentially spurious.

V. CONCLUSION

This project demonstrates how data mining techniques can be used to discover meaningful patterns in government economic data. By emphasizing interpretation and critical assessment over prediction, the project aligns with the core goals of data mining and the KDD process.